```
Q = [2 2; 2 6]
```

```
Q = 2×2
     2     2
     2     6
```

```
C = [2; 10]
```

```
C = 2×1
     2
    10
```

```
pinv(Q)*Q
```

```
ans = 2×2
    1.0000    0.0000
    0.0000    1.0000
```

```
x_min = -pinv(Q) * C
```

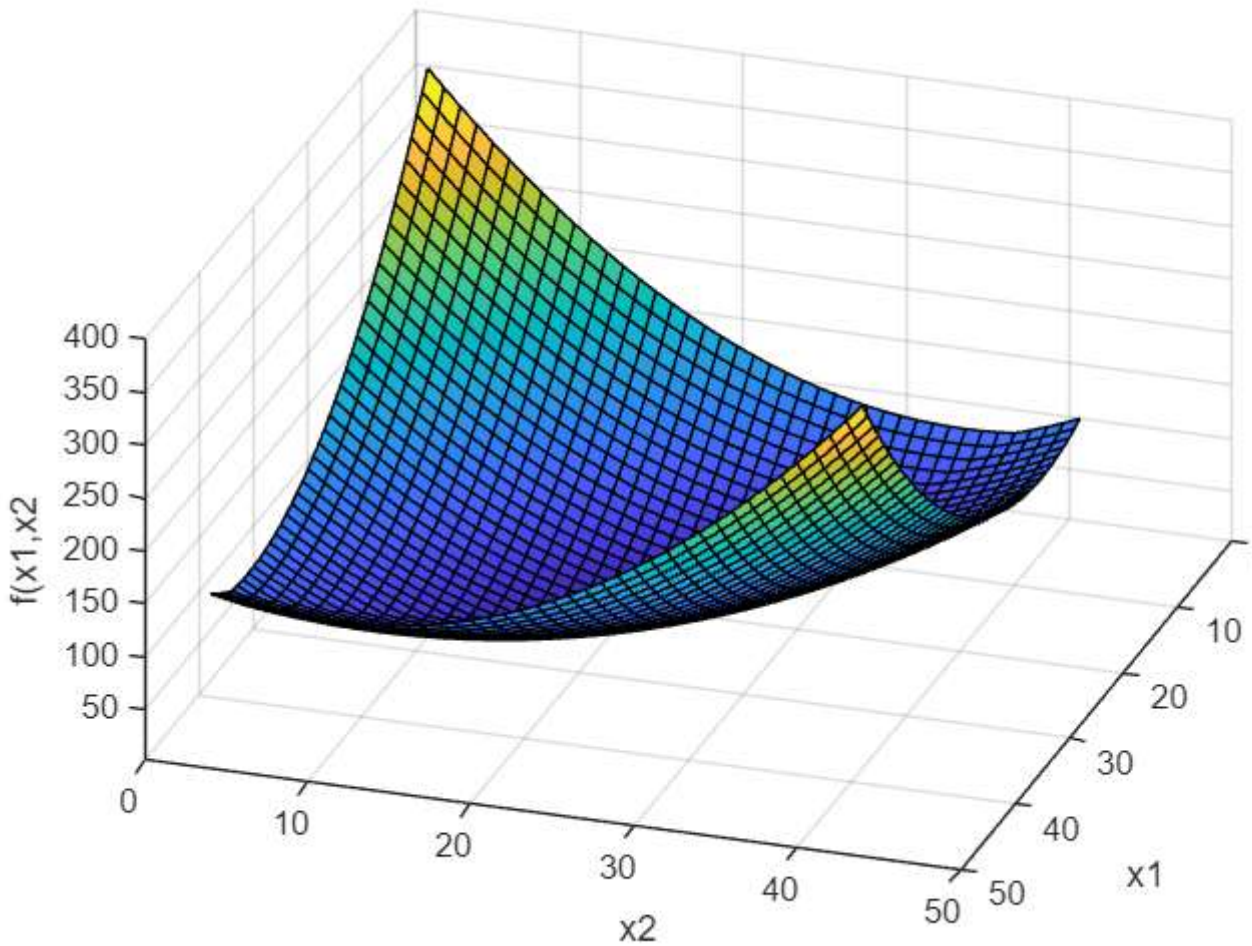
```
x_min = 2×1
    1.0000
   -2.0000
```

```
rank(Q)
```

```
ans = 2
```

```
%Starting here, you can define a function f(x1,x2) and plot it. If you want
%a heatmap, change the "surf(f)" command to "map(f)"

x1 = -10:.5:10;
x2 = -10:.5:10;

clear min1
for i = 1:size(x1,2)
    for ii = 1:size(x2,2)
    f(i,ii) = x1(i)^2 + x1(i)*x2(ii) +x2(ii)^2 + 3*abs(x1(i) + x2(ii) + 2);
        if exist("min","var") ~= 1
            min1 = [i, ii, f(i,ii)];
        end
        if f(i,ii) < min(3)
            min1 = [i, ii, f(i,ii)];
        end
    end
end
```

```
hold on
surf(f)
view([119 38])
xlabel("x1")
ylabel("x2")
zlabel("f(x1,x2")
grid("on")

plot3(min1(1), min1(2), min1(3));

hold off
```



HW2 Problem3, Jet Thrust

```
syms rho S C_D0 K W T V
res = 1;

T = 0.5*rho*S*V^2*C_D0 + (K*W^2)/ (0.5*rho*S*V^2)
```

T =

$$\frac{C_{D0}\, S\, V^2\, \rho}{2} + \frac{2\, K\, W^2}{S\, V^2\, \rho}$$

```
T0 = subs(T,[W S C_D0 K rho], [73000 950 0.015 0.05 1.7553e-3])
```

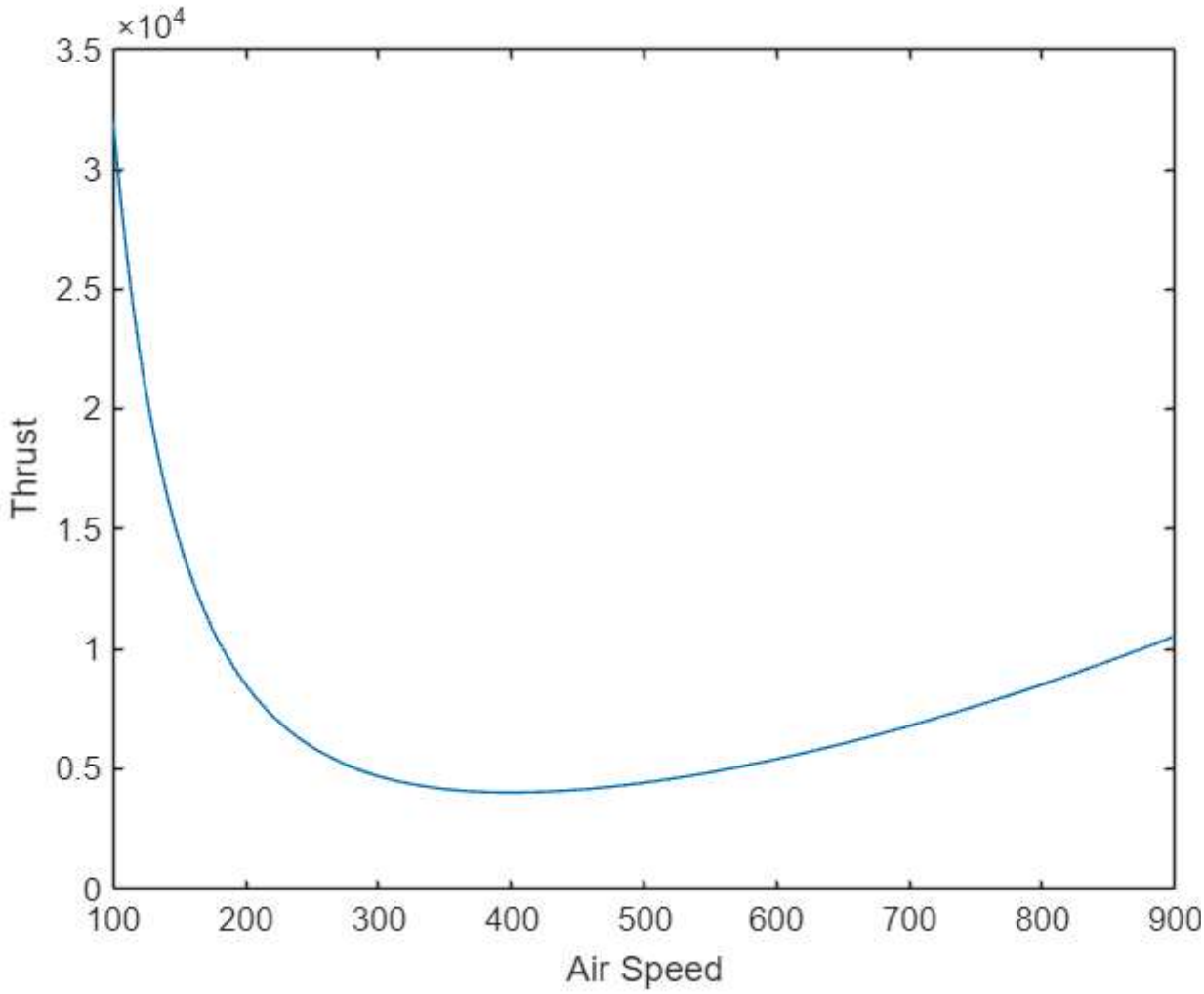
T0 =

$$\frac{2457567479219955014041600 0}{76901478447383143\,V^2} + \frac{230704435342149429\,V^2}{18446744073709551616}$$

```
V0 = 100:res:900;
T0_V0 = double(subs(T0, V, V0))
```

```
T0_V0 = 1×801
10^4 ×
    3.2082    3.1455    3.0847    3.0256    2.9682    2.9124    2.8582    2.8056    2.7544    2.70
```

```
plot(V0,T0_V0)
xlabel("Air Speed")
ylabel("Thrust")
```




```
min2 = [min(V0), min(T0_V0)]
```

```
min2 = 1×2
10^3 ×
    0.1000    3.9984
```